



EXAM PAPERS PRACTICE

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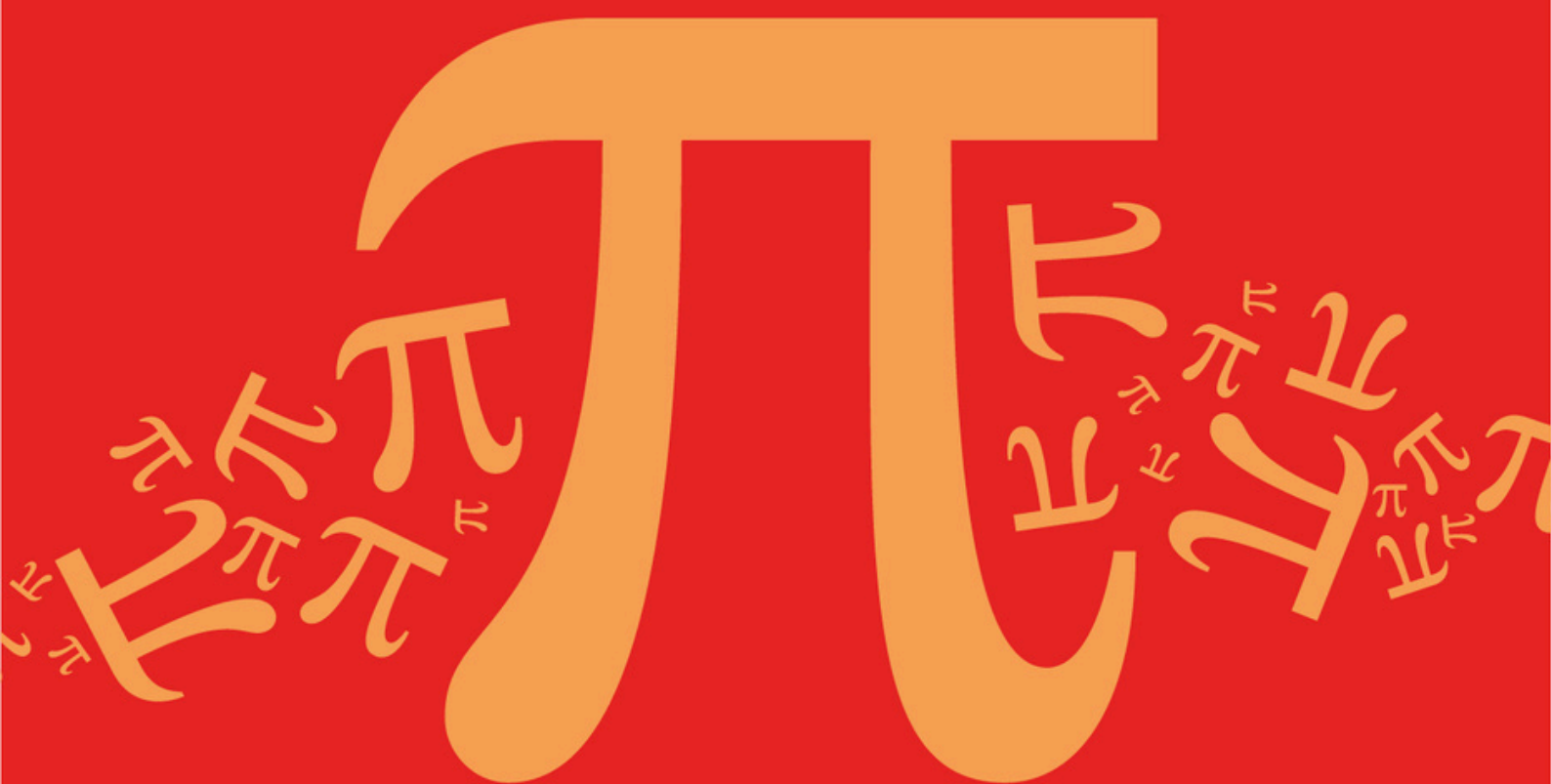
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Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D2: Transform Polynomial Equation

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

D: Further Algebra and Functions

Sub topic

D2: Transform Polynomial Equation

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

The roots of the quadratic equation

$$x^2 + 2x - 5 = 0$$

are α and β

- (a) Write down the value of $\alpha + \beta$ and the value of $\alpha\beta$.

(2)

- (b) Calculate the value of $\alpha^2 + \beta^2$.

(2)

- (c) Find a quadratic equation which has roots $\alpha^3\beta + 1$ and $\alpha\beta^3 + 1$.

(5)

(Total 9 marks)

Q2.

The equation

$$2x^2 + 3x - 6 = 0$$

has roots α and β .

- (a) Write down the value of $\alpha + \beta$ and the value of $\alpha\beta$.

(2)

- (b) Hence show that $\alpha^3 + \beta^3 = -\frac{135}{8}$.

(3)

- (c) Find a quadratic equation, with integer coefficients, whose roots are $\alpha + \frac{\alpha}{\beta^2}$ and $\beta + \frac{\beta}{\alpha^2}$.

(6)

(Total 11 marks)

Q3.

The quadratic equation

$$2x^2 + 7x + 8 = 0$$

has roots α and β .

- (a) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

- (b) Show that $\alpha^2 + \beta^2 = \frac{17}{4}$.

(2)

- (c) Find a quadratic equation, with integer coefficients, which has roots

$$\frac{1}{\alpha^2} \text{ and } \frac{1}{\beta^2}$$

(5)

(Total 9 marks)

Q4.

The quadratic equation

$$5x^2 - 7x + 1 = 0$$

has roots α and β .

- (a) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

- (b) Show that $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{39}{5}$.

(3)

- (c) Find a quadratic equation, with integer coefficients, which has roots

$$\alpha + \frac{1}{\alpha} \text{ and } \beta + \frac{1}{\beta}$$

(5)

(Total 10 marks)

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Q5.

The cubic equation

$$z^3 + pz + q = 0$$

has roots α , β and γ .

- (a) (i) Write down the value of $\alpha + \beta + \gamma$.

(1)

- (ii) Express $\alpha\beta\gamma$ in terms of q .

(1)

- (b) Show that

$$\alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma$$

(3)

- (c) Given that $\alpha = 4 + 7i$ and that p and q are real, find the values of :

- (i) β and γ ;

(2)

(ii) p and q .

(3)

(d) Find a cubic equation with integer coefficients which has roots, $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$.

(3)

(Total 13 marks)

Q6.

The quadratic equation $x^2 - 6x + 18 = 0$ has roots α and β .

(a) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

(b) Find a quadratic equation, with integer coefficients, which has roots α^2 and β^2 .

(4)

(c) Hence find the values of α^2 and β^2 .

(1)

(Total 7 marks)

Q7.

The equation

$$4x^2 + 6x + 3 = 0$$

has roots a and β .

(a) Write down the values of $a + \beta$ and $a\beta$.

(2)

(b) Show that $\alpha^2 + \beta^2 = \frac{3}{4}$.

(2)

(c) Find an equation, with integer coefficients, which has roots

$$3a - \beta \text{ and } 3\beta - a$$

(5)

(Total 9 marks)

Q8.

The quadratic equation

$$3x^2 - 6x + 1 = 0$$

has roots α and β .

(a) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

- (b) Show that $\alpha^3 + \beta^3 = 6$.

(3)

- (c) Find a quadratic equation, with integer coefficients, which has roots $\frac{\alpha^2}{\beta}$ and $\frac{\beta^2}{\alpha}$.

(4)

(Total 9 marks)

Q9.

The quadratic equation

$$x^2 - 4x + 10 = 0$$

has roots α and β .

- (a) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

- (b) Show that $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{2}{5}$.

(2)

- (c) Find a quadratic equation, with integer coefficients, which has roots $\alpha + \frac{2}{\beta}$ and $\beta + \frac{2}{\alpha}$.

(6)

(Total 10 marks)

Q10.

The equation

$$2x^2 + x - 8 = 0$$

has roots α and β .

- (a) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

- (b) Find the value of $\alpha^2 + \beta^2$.

(2)

- (c) Find a quadratic equation which has roots $4\alpha^2$ and $4\beta^2$. Give your answer in the form $x^2 + px + q = 0$, where p and q are integers.

(3)

(Total 7 marks)

Q11.

- (a) (i) It is given that α and β are the roots of the equation

$$x^2 - 2x + 4 = 0$$

Without solving this equation, show that α^3 and β^3 are the roots of the equation

$$x^2 + 16x + 64 = 0$$

(6)

- (ii) State, giving a reason, whether the roots of the equation

$$x^2 + 16x + 64 = 0$$

are real and equal, real and distinct, or non-real.

(2)

- (b) Solve the equation

$$x^2 - 2x + 4 = 0$$

(2)

- (c) Use your answers to parts (a) and (b) to show that

$$(1 + i\sqrt{3})^3 = (1 - i\sqrt{3})^3$$

(2)

(Total 12 marks)

Q12.

The cubic equation

$$z^3 + iz^2 + 3z - (1 + i) = 0$$

has roots α , β and γ .

- (a) Write down the value of:

(i) $\alpha + \beta + \gamma$;

(1)

(ii) $\alpha\beta + \beta\gamma + \gamma\alpha$;

(1)

(iii) $\alpha\beta\gamma$.

(1)

- (b) Find the value of:

(i) $\alpha^2 + \beta^2 + \gamma^2$;

(3)

(ii) $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$;

(4)

(iii) $\alpha^2\beta^2\gamma^2$.

(2)

(c) Hence write down a cubic equation whose roots are α^2 , β^2 and γ^2 .

(2)

(Total 14 marks)

Q13.

The equation

$$x^2 + x + 5 = 0$$

has roots α and β .

(a) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

(b) Find the value of $\alpha^2 + \beta^2$.

(2)

(c) Show that $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = -\frac{9}{5}$.

(2)

(d) Find a quadratic equation, with integer coefficients, which has roots $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$.

(2)

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Q14.

The quadratic equation

$$2x^2 - x + 4 = 0$$

has roots α and β .

(a) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

(b) Show that $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{1}{4}$.

(2)

(c) Find a quadratic equation with integer coefficients such that the roots of the equation are $\frac{4}{\alpha}$ and $\frac{4}{\beta}$.

(3)

(Total 7 marks)

Q15.

The quadratic equation

$$3x^2 - 6x + 2 = 0$$

has roots α and β .

- (a) Write down the numerical values of $\alpha + \beta$ and $\alpha\beta$.

(2)

- (b) (i) Expand $(\alpha + \beta)^3$.

(1)

- (ii) Show that $\alpha^3 + \beta^3 = 4$.

(3)

- (c) Find a quadratic equation with roots α^3 and β^3 , giving your answer in the form $px^2 + qx + r = 0$, where p , q and r are integers.

(3)

(Total 9 marks)

Q16.

The equation

$$x^2 - 5x - 2 = 0$$

has roots α and β .

- (a) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

- (b) Find the value of $\alpha^2\beta + \alpha\beta^2$.

(2)

- (c) Find a quadratic equation which has roots

$$\alpha^2\beta \text{ and } \alpha\beta^2$$

(3)

(Total 7 marks)

Q17.

The equation

$$x^2 - 4x + 13 = 0$$

has roots α and β .

- (a) (i) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

- (ii) Deduce that $\alpha^2 + \beta^2 = -10$.

- (2)
- (iii) Explain why the statement $\alpha^2 + \beta^2 = -10$ implies that α and β cannot both be real.
- (2)
- (b) Find in the form $p + iq$ the values of:
- (i) $(\alpha + i) + (\beta + i)$;
- (1)
- (ii) $(\alpha + i)(\beta + i)$.
- (2)
- (c) Hence find a quadratic equation with roots $(\alpha + i)$ and $(\beta + i)$.
- (2)

(Total 11 marks)